

```
# **BANK DATA CENTER - COMPLETE CONFIGURATIONS**
```

```
## **ALL DEVICE CONFIGURATIONS - COPY & PASTE READY**
```

```
---
```

```
# **CORE ROUTERS**
```

```
## **CORE-R1 Configuration**
```

```
```
```

```
configure terminal
```

```
hostname CORE-R1
```

```
! ===== WAN INTERFACES (Public IPs) =====
```

```
interface GigabitEthernet0/0
```

```
description Link to ISP-R1 (AS 100)
```

```
ip address 203.0.113.2 255.255.255.252
```

```
ip nat outside
```

```
ip ospf message-digest-key 1 md5 BANKOSPF2026
```

```
no shutdown
```

```
exit
```

```
interface GigabitEthernet0/1
```

```
description Link to ISP-R2 (AS 200)
```

```
ip address 203.0.113.6 255.255.255.252
```

```
ip nat outside
```

```
ip ospf message-digest-key 1 md5 BANKOSPF2026
```

```
no shutdown
```

```
exit
```

```
! ===== OSPF ROUTING =====
```

```
router ospf 1
```

```
router-id 1.1.1.3
```

```
network 10.0.0.0 0.0.255.255 area 0
```

```
area 0 authentication message-digest
```

```
exit
```

```
! ===== BGP (Internet Connectivity) =====
```

```
router bgp 65001
```

```
bgp router-id 1.1.1.3
```

```
neighbor 203.0.113.1 remote-as 100
neighbor 203.0.113.5 remote-as 200
redistribute ospf 1 metric 0
redistribute connected
```

```
address-family ipv4
 neighbor 203.0.113.1 activate
 neighbor 203.0.113.5 activate
exit-address-family
exit
```

```
! ===== NAT CONFIGURATION =====
```

```
! Static NAT for Load-Balancer (always maps to public IP)
```

```
ip nat inside source static 10.0.20.100 203.0.113.150
```

```
! Dynamic NAT/PAT for all internal hosts
```

```
access-list INSIDE_HOSTS permit 10.0.0.0 0.0.255.255
```

```
ip nat inside source list INSIDE_HOSTS interface GigabitEthernet0/0 overload
```

```
! ===== EXTENDED ACL (Block RFC1918 spoofing from internet) =====
```

```
access-list BLOCK_INTERNET_IN extended deny ip any 10.0.0.0 255.255.0.0
```

```
access-list BLOCK_INTERNET_IN extended deny ip any 172.16.0.0 255.240.0.0
```

```
access-list BLOCK_INTERNET_IN extended deny ip any 192.168.0.0 255.255.0.0
```

```
access-list BLOCK_INTERNET_IN extended permit ip any any
```

```
access-list 101 permit ip any any
```

```
! ===== QoS CONFIGURATION =====
```

```
class-map match-any CRITICAL
```

```
 match protocol http
```

```
 match protocol https
```

```
exit
```

```
class-map match-any HIGH
```

```
 match dscp af11 af12 af13
```

```
exit
```

```
class-map match-any NORMAL
```

```
 match dscp default
```

```
exit
```

```

policy-map BANK_QoS
 class CRITICAL
 priority percent 40
 exit
 class HIGH
 bandwidth percent 30
 exit
 class NORMAL
 bandwidth percent 20
 exit
 class-default
 bandwidth percent 10
 exit
exit

interface GigabitEthernet0/0
 service-policy output BANK_QoS
exit

interface GigabitEthernet0/1
 service-policy output BANK_QoS
exit

end
write memory
```


---



## **CORE-R2 Configuration**



```



```

configure terminal
hostname CORE-R2

! ===== WAN INTERFACES =====
interface GigabitEthernet0/0
  description Link to ISP-R1 (AS 100)
  ip address 203.0.113.3 255.255.255.252
  ip nat outside
  ip ospf message-digest-key 1 md5 BANKOSPF2026

```


```

```
no shutdown
exit
```

```
interface GigabitEthernet0/1
description Link to ISP-R2 (AS 200)
ip address 203.0.113.7 255.255.255.252
ip nat outside
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
! ===== OSPF ROUTING =====
```

```
router ospf 1
router-id 2.2.2.3
network 10.0.0.0 0.0.255.255 area 0
area 0 authentication message-digest
exit
```

```
! ===== BGP =====
```

```
router bgp 65001
bgp router-id 2.2.2.3
neighbor 203.0.113.1 remote-as 100
neighbor 203.0.113.5 remote-as 200
redistribute ospf 1 metric 0
redistribute connected
```

```
address-family ipv4
neighbor 203.0.113.1 activate
neighbor 203.0.113.5 activate
exit-address-family
exit
```

```
! ===== NAT CONFIGURATION =====
```

```
ip nat inside source static 10.0.20.100 203.0.113.150
```

```
access-list INSIDE_HOSTS permit 10.0.0.0 0.0.255.255
ip nat inside source list INSIDE_HOSTS interface GigabitEthernet0/0 overload
```

```
! ===== QoS =====
```

```
policy-map BANK_QoS
class CRITICAL
```

```

 priority percent 40
exit
class HIGH
 bandwidth percent 30
exit
class NORMAL
 bandwidth percent 20
exit
class-default
 bandwidth percent 10
exit
exit

interface GigabitEthernet0/0
 service-policy output BANK_QoS
exit

interface GigabitEthernet0/1
 service-policy output BANK_QoS
exit

end
write memory
```


---



```

ISP ROUTERS

ISP-R1 Configuration (AS 100)

```
configure terminal
hostname ISP-R1

interface GigabitEthernet0/0
    description Link to CORE-R1
    ip address 203.0.113.1 255.255.255.252
    no shutdown
exit

```


```

```
interface GigabitEthernet0/1
description Link to CORE-R2
ip address 203.0.113.1 255.255.255.252
no shutdown
exit
```

```
end
write memory
\ \
```

---

```
ISP-R2 Configuration (AS 200)
```

```
\ \
```

```
configure terminal
hostname ISP-R2
```

```
interface GigabitEthernet0/0
description Link to CORE-R1
ip address 203.0.113.5 255.255.255.252
no shutdown
exit
```

```
interface GigabitEthernet0/1
description Link to CORE-R2
ip address 203.0.113.5 255.255.255.252
no shutdown
exit
```

```
end
write memory
\ \
```

---

```
CORE LAYER L3 SWITCHES
```

```
CORE-SW1 Configuration (HSRP ACTIVE - Priority 110)
```

```
\ \
```

```
configure terminal
hostname CORE-SW1

ip routing

! ===== VLAN CREATION =====
vlan 10
 name MANAGEMENT
exit
vlan 20
 name SERVERS
exit
vlan 30
 name DATABASE
exit
vlan 40
 name INTERNAL
exit
vlan 50
 name SOC
exit
vlan 60
 name SIEM
exit
vlan 70
 name STORAGE
exit
vlan 80
 name PEERING
exit
vlan 99
 name BLACKHOLE
exit

! ===== TRUNK PORTS TO FIREWALLS =====
interface GigabitEthernet1/1
 description Trunk to FW1
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
```

```
exit
```

```
interface GigabitEthernet1/2
description Trunk to FW2
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit
```

```
! ===== TRUNK PORTS TO AGGREGATION =====
```

```
interface GigabitEthernet2/1
description Trunk to AGG-SW1
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit
```

```
interface GigabitEthernet2/2
description Trunk to AGG-SW2
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit
```

```
! ===== VLAN SVI INTERFACES (Layer 3 Gateways) =====
```

```
interface Vlan10
description Management VLAN Gateway
ip address 10.0.10.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan20
description Servers VLAN Gateway
ip address 10.0.20.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```



```
interface Vlan30
description Database VLAN Gateway
ip address 10.0.30.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan40
description Internal VLAN Gateway
ip address 10.0.40.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan50
description SOC VLAN Gateway
ip address 10.0.50.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan60
description SIEM VLAN Gateway
ip address 10.0.60.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan70
description Storage VLAN Gateway
ip address 10.0.70.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan80
description Peering VLAN Gateway
ip address 10.0.80.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
```

exit

! ===== OSPF CONFIGURATION =====

router ospf 1

router-id 3.3.3.1

network 10.0.0.0 0.0.255.255 area 0

area 0 authentication message-digest

passive-interface Vlan10

passive-interface Vlan20

passive-interface Vlan30

passive-interface Vlan40

passive-interface Vlan50

passive-interface Vlan60

passive-interface Vlan70

passive-interface Vlan80

exit

! ===== HSRP CONFIGURATION (ACTIVE - Priority 110) =====

interface Vlan10

standby 1 ip 10.0.10.254

standby 1 priority 110

standby 1 preempt

exit

interface Vlan20

standby 2 ip 10.0.20.254

standby 2 priority 110

standby 2 preempt

exit

interface Vlan30

standby 3 ip 10.0.30.254

standby 3 priority 110

standby 3 preempt

exit

interface Vlan40

standby 4 ip 10.0.40.254

standby 4 priority 110

standby 4 preempt

exit

```
interface Vlan50
 standby 5 ip 10.0.50.254
 standby 5 priority 110
 standby 5 preempt
exit
```

```
interface Vlan60
 standby 6 ip 10.0.60.254
 standby 6 priority 110
 standby 6 preempt
exit
```

```
interface Vlan70
 standby 7 ip 10.0.70.254
 standby 7 priority 110
 standby 7 preempt
exit
```

```
interface Vlan80
 standby 8 ip 10.0.80.254
 standby 8 priority 110
 standby 8 preempt
exit
```

```
end
write memory
\ \ \
```

---

```
CORE-SW2 Configuration (HSRP STANDBY - Priority 100)
```

```
\ \ \
```

```
configure terminal
hostname CORE-SW2
```

```
ip routing
```

```
! ===== VLAN CREATION (Same as CORE-SW1) =====
```

```
vlan 10
```

```
name MANAGEMENT
exit
vlan 20
name SERVERS
exit
vlan 30
name DATABASE
exit
vlan 40
name INTERNAL
exit
vlan 50
name SOC
exit
vlan 60
name SIEM
exit
vlan 70
name STORAGE
exit
vlan 80
name PEERING
exit
vlan 99
name BLACKHOLE
exit
```

```
! ===== TRUNK PORTS =====
```

```
interface GigabitEthernet1/1
description Trunk to FW1
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit
```

```
interface GigabitEthernet1/2
description Trunk to FW2
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
```

```
no shutdown
exit
```

```
interface GigabitEthernet2/1
description Trunk to AGG-SW1
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit
```

```
interface GigabitEthernet2/2
description Trunk to AGG-SW2
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit
```

! ===== VLAN SVI INTERFACES (Same IPs as CORE-SW1) =====

```
interface Vlan10
ip address 10.0.10.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan20
ip address 10.0.20.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan30
ip address 10.0.30.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan40
ip address 10.0.40.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
```

```
no shutdown
exit
```

```
interface Vlan50
ip address 10.0.50.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan60
ip address 10.0.60.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan70
ip address 10.0.70.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan80
ip address 10.0.80.1 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
! ===== OSPF =====
router ospf 1
router-id 3.3.3.2
network 10.0.0.0 0.0.255.255 area 0
area 0 authentication message-digest
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan50
passive-interface Vlan60
passive-interface Vlan70
passive-interface Vlan80
exit
```

! ===== HSRP (STANDBY - Priority 100) =====

```
interface Vlan10
 standby 1 ip 10.0.10.254
 standby 1 priority 100
 standby 1 preempt
exit
```

```
interface Vlan20
 standby 2 ip 10.0.20.254
 standby 2 priority 100
 standby 2 preempt
exit
```

```
interface Vlan30
 standby 3 ip 10.0.30.254
 standby 3 priority 100
 standby 3 preempt
exit
```

```
interface Vlan40
 standby 4 ip 10.0.40.254
 standby 4 priority 100
 standby 4 preempt
exit
```

```
interface Vlan50
 standby 5 ip 10.0.50.254
 standby 5 priority 100
 standby 5 preempt
exit
```

```
interface Vlan60
 standby 6 ip 10.0.60.254
 standby 6 priority 100
 standby 6 preempt
exit
```

```
interface Vlan70
 standby 7 ip 10.0.70.254
 standby 7 priority 100
```

```
standby 7 preempt
exit
```

```
interface Vlan80
standby 8 ip 10.0.80.254
standby 8 priority 100
standby 8 preempt
exit
```

```
end
write memory
\ \
```

```

```

```
AGGREGATION LAYER L3 SWITCHES
```

```
AGG-SW1 Configuration (HSRP ACTIVE on VLAN 20 - Priority 110)
```

```
\ \
```

```
configure terminal
hostname AGG-SW1
```

```
ip routing
```

```
! ===== VLAN CREATION =====
```

```
vlan 10
name MANAGEMENT
exit
vlan 20
name SERVERS
exit
vlan 30
name DATABASE
exit
vlan 40
name INTERNAL
exit
vlan 50
name SOC
exit
```



```
vlan 60
 name SIEM
exit
vlan 70
 name STORAGE
exit
vlan 80
 name PEERING
exit
vlan 99
 name BLACKHOLE
exit
```

```
! ===== TRUNK PORTS TO CORE =====
```

```
interface GigabitEthernet1/1
 description Trunk to CORE-SW1
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit
```

```
interface GigabitEthernet1/2
 description Trunk to CORE-SW2
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit
```

```
! ===== TRUNK PORTS TO ACCESS SWITCHES =====
```

```
interface range FastEthernet0/1-6
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit
```

```
! ===== VLAN SVI INTERFACES (.2 addresses) =====
```

```
interface Vlan10
 ip address 10.0.10.2 255.255.255.0
```

```
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan20
ip address 10.0.20.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan30
ip address 10.0.30.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan40
ip address 10.0.40.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan50
ip address 10.0.50.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan60
ip address 10.0.60.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan70
ip address 10.0.70.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan80
```

```
ip address 10.0.80.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
! ===== OSPF =====
router ospf 1
router-id 5.5.5.1
network 10.0.0.0 0.0.255.255 area 0
area 0 authentication message-digest
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan50
passive-interface Vlan60
passive-interface Vlan70
passive-interface Vlan80
exit
```

```
! ===== HSRP (Secondary level on VLAN 20 only) =====
interface Vlan20
standby 20 ip 10.0.20.253
standby 20 priority 110
standby 20 preempt
exit
```

```
end
write memory
^^
```

---

```
AGG-SW2 Configuration (HSRP STANDBY on VLAN 20 - Priority 100)
```

```
^^
configure terminal
hostname AGG-SW2
```

```
ip routing
```

! ===== VLAN CREATION =====

```
vlan 10
 name MANAGEMENT
exit
vlan 20
 name SERVERS
exit
vlan 30
 name DATABASE
exit
vlan 40
 name INTERNAL
exit
vlan 50
 name SOC
exit
vlan 60
 name SIEM
exit
vlan 70
 name STORAGE
exit
vlan 80
 name PEERING
exit
vlan 99
 name BLACKHOLE
exit
```

! ===== TRUNK PORTS TO CORE =====

```
interface GigabitEthernet1/1
 description Trunk to CORE-SW1
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit
```

```
interface GigabitEthernet1/2
 description Trunk to CORE-SW2
 switchport mode trunk
```

```
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit
```

! ===== TRUNK PORTS TO ACCESS SWITCHES =====

```
interface range FastEthernet0/1-6
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit
```

! ===== VLAN SVI INTERFACES =====

```
interface Vlan10
ip address 10.0.10.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan20
ip address 10.0.20.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan30
ip address 10.0.30.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan40
ip address 10.0.40.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan50
ip address 10.0.50.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
```

```
no shutdown
exit
```

```
interface Vlan60
ip address 10.0.60.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan70
ip address 10.0.70.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
interface Vlan80
ip address 10.0.80.2 255.255.255.0
ip ospf message-digest-key 1 md5 BANKOSPF2026
no shutdown
exit
```

```
! ===== OSPF =====
router ospf 1
router-id 5.5.5.2
network 10.0.0.0 0.0.255.255 area 0
area 0 authentication message-digest
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan50
passive-interface Vlan60
passive-interface Vlan70
passive-interface Vlan80
exit
```

```
! ===== HSRP (Standby on VLAN 20) =====
interface Vlan20
standby 20 ip 10.0.20.253
standby 20 priority 100
standby 20 preempt
```

```
exit

end
write memory
\ \ \

ACCESS LAYER L2 SWITCHES

ACC-SW1 Configuration

\ \ \

configure terminal
hostname ACC-SW1

! ===== VLAN CREATION =====
vlan 10
 name MANAGEMENT
exit
vlan 20
 name SERVERS
exit
vlan 30
 name DATABASE
exit
vlan 40
 name INTERNAL
exit
vlan 50
 name SOC
exit
vlan 60
 name SIEM
exit
vlan 70
 name STORAGE
exit
vlan 80
 name PEERING
exit
```

```

vlan 99
 name BLACKHOLE
exit

! ===== TRUNK PORTS TO AGGREGATION =====
interface GigabitEthernet0/1
 description Trunk to AGG-SW1
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit

interface GigabitEthernet0/2
 description Trunk to AGG-SW2
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit

! ===== SHUTDOWN UNUSED PORTS =====
interface range FastEthernet0/3-24
 switchport access vlan 99
 shutdown
exit

end
write memory
\ \ \

ACC-SW2 Configuration

\ \ \

configure terminal
hostname ACC-SW2

vlan 10
 name MANAGEMENT

```



```
exit
vlan 20
 name SERVERS
exit
vlan 30
 name DATABASE
exit
vlan 40
 name INTERNAL
exit
vlan 50
 name SOC
exit
vlan 60
 name SIEM
exit
vlan 70
 name STORAGE
exit
vlan 80
 name PEERING
exit
vlan 99
 name BLACKHOLE
exit
```

```
interface GigabitEthernet0/1
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit
```

```
interface GigabitEthernet0/2
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit
```

```
interface range FastEthernet0/3-24
```

```
switchport access vlan 99
shutdown
exit
```

```
end
write memory
\ \ \
```

```

```

```
ACC-SW3 Configuration
```

```
\ \ \
```

```
configure terminal
hostname ACC-SW3
```

```
vlan 10
name MANAGEMENT
```

```
exit
```

```
vlan 20
name SERVERS
```

```
exit
```

```
vlan 30
name DATABASE
```

```
exit
```

```
vlan 40
name INTERNAL
```

```
exit
```

```
vlan 50
name SOC
```

```
exit
```

```
vlan 60
name SIEM
```

```
exit
```

```
vlan 70
name STORAGE
```

```
exit
```

```
vlan 80
name PEERING
```

```
exit
```

```
vlan 99
```

```
name BLACKHOLE
exit

interface GigabitEthernet0/1
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit

interface GigabitEthernet0/2
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit

interface range FastEthernet0/3-24
switchport access vlan 99
shutdown
exit

end
write memory
```
---

## **ACC-SW4 Configuration**

```
name ACC-SW4

configure terminal
hostname ACC-SW4

vlan 10
name MANAGEMENT
exit
vlan 20
name SERVERS
exit
vlan 30

```

```
 name DATABASE
exit
vlan 40
 name INTERNAL
exit
vlan 50
 name SOC
exit
vlan 60
 name SIEM
exit
vlan 70
 name STORAGE
exit
vlan 80
 name PEERING
exit
vlan 99
 name BLACKHOLE
exit

interface GigabitEthernet0/1
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit

interface GigabitEthernet0/2
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit

interface range FastEthernet0/3-24
 switchport access vlan 99
 shutdown
exit

end
```

```
write memory
```

```
^^^
```

```

```

```
ACC-SW5 Configuration
```

```
^^^
```

```
configure terminal
```

```
hostname ACC-SW5
```

```
vlan 10
```

```
name MANAGEMENT
```

```
exit
```

```
vlan 20
```

```
name SERVERS
```

```
exit
```

```
vlan 30
```

```
name DATABASE
```

```
exit
```

```
vlan 40
```

```
name INTERNAL
```

```
exit
```

```
vlan 50
```

```
name SOC
```

```
exit
```

```
vlan 60
```

```
name SIEM
```

```
exit
```

```
vlan 70
```

```
name STORAGE
```

```
exit
```

```
vlan 80
```

```
name PEERING
```

```
exit
```

```
vlan 99
```

```
name BLACKHOLE
```

```
exit
```

```
interface GigabitEthernet0/1
```

```
switchport mode trunk
```

```
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit
```

```
interface GigabitEthernet0/2
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
no shutdown
exit
```

```
interface range FastEthernet0/3-24
switchport access vlan 99
shutdown
exit
```

```
end
write memory
\`\`\`
```

---

```
ACC-SW6 Configuration
```

```
\`\`\`
```

```
configure terminal
hostname ACC-SW6
```

```
vlan 10
name MANAGEMENT
exit
vlan 20
name SERVERS
exit
vlan 30
name DATABASE
exit
vlan 40
name INTERNAL
exit
```

```
vlan 50
 name SOC
exit
vlan 60
 name SIEM
exit
vlan 70
 name STORAGE
exit
vlan 80
 name PEERING
exit
vlan 99
 name BLACKHOLE
exit

interface GigabitEthernet0/1
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit

interface GigabitEthernet0/2
 switchport mode trunk
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20,30,40,50,60,70,80,99
 no shutdown
exit

interface range FastEthernet0/3-24
 switchport access vlan 99
 shutdown
exit

end
write memory
\ \

```

```
FIREWALLS (ASA 5506-X)

FW1 Configuration (Outside-to-DMZ)

'''

configure terminal
hostname FW1

! ===== INTERFACES =====
interface GigabitEthernet1/1
 nameif outside
 security-level 0
 ip address 203.0.113.2 255.255.255.252
 no shutdown
exit

interface GigabitEthernet1/2
 nameif dmz
 security-level 50
 ip address 10.0.80.1 255.255.255.0
 no shutdown
exit

! ===== OSPF ROUTING =====
router ospf 1
 router-id 1.1.1.4
exit

! ===== ACL: Block RFC1918 from Internet =====
access-list FW1_IN extended deny ip any 10.0.0.0 255.255.0.0
access-list FW1_IN extended deny ip any 172.16.0.0 255.240.0.0
access-list FW1_IN extended deny ip any 192.168.0.0 255.255.0.0
access-list FW1_IN extended permit tcp any any eq 80
access-list FW1_IN extended permit tcp any any eq 443
access-list FW1_IN extended permit tcp any any established
access-list FW1_IN extended deny ip any any

access-group FW1_IN in interface outside

write memory
'''
```



---

## \*\*FW2 Configuration (DMZ-to-Inside)\*\*

```

configure terminal

hostname FW2

! ===== INTERFACES =====

interface GigabitEthernet1/1

nameif dmz

security-level 50

ip address 10.0.80.2 255.255.255.0

no shutdown

exit

interface GigabitEthernet1/2

nameif inside

security-level 100

ip address 10.0.50.254 255.255.255.0

no shutdown

exit

! ===== OSPF ROUTING =====

router ospf 1

router-id 2.2.2.4

exit

! ===== ACL: Database Access Control =====

access-list FW2_DB extended permit tcp 10.0.20.0 255.255.255.0 10.0.30.0 255.255.255.0 eq 3306

access-list FW2_DB extended permit tcp 10.0.50.0 255.255.255.0 10.0.30.0 255.255.255.0 eq 3306

access-list FW2_DB extended deny ip any any

access-group FW2_DB in interface inside

write memory

```

---

# \*\*SERVER CONFIGURATIONS\*\*

## \*\*Server-PT IP Configuration Commands\*\*

These are entered in the Server-PT terminal (not traditional CLI):

\*\*Load-Balancer (VLAN 20):\*\*

\\

#IPCONFIG: STATIC, 10.0.20.100, 255.255.255.0, 10.0.20.254

\\

\*\*DB-SERVER (VLAN 30):\*\*

\\

#IPCONFIG: STATIC, 10.0.30.100, 255.255.255.0, 10.0.30.254

\\

\*\*SOC (VLAN 50):\*\*

\\

#IPCONFIG: STATIC, 10.0.50.50, 255.255.255.0, 10.0.50.254

\\

\*\*SIEM (VLAN 60):\*\*

\\

#IPCONFIG: STATIC, 10.0.60.50, 255.255.255.0, 10.0.60.254

\\

\*\*IDS (VLAN 50):\*\*

\\

#IPCONFIG: STATIC, 10.0.50.51, 255.255.255.0, 10.0.50.254

\\

\*\*IPS (VLAN 50):\*\*

\\

#IPCONFIG: STATIC, 10.0.50.52, 255.255.255.0, 10.0.50.254

\\

---

# \*\*QUICK REFERENCE - IP ADDRESSING\*\*

## \*\*Router WAN IPs\*\*

```

CORE-R1 Gi0/0: 203.0.113.2/30 (to ISP-R1)

CORE-R1 Gi0/1: 203.0.113.6/30 (to ISP-R2)

CORE-R2 Gi0/0: 203.0.113.3/30 (to ISP-R1)

CORE-R2 Gi0/1: 203.0.113.7/30 (to ISP-R2)

ISP-R1 Gi0/0: 203.0.113.1/30

ISP-R1 Gi0/1: 203.0.113.1/30

ISP-R2 Gi0/0: 203.0.113.5/30

ISP-R2 Gi0/1: 203.0.113.5/30

```

## \*\*VLAN SVI IPs (Core)\*\*

```

VLAN 10 (MANAGEMENT): 10.0.10.1/24 → VIP: 10.0.10.254

VLAN 20 (SERVERS): 10.0.20.1/24 → VIP: 10.0.20.254

VLAN 30 (DATABASE): 10.0.30.1/24 → VIP: 10.0.30.254

VLAN 40 (INTERNAL): 10.0.40.1/24 → VIP: 10.0.40.254

VLAN 50 (SOC): 10.0.50.1/24 → VIP: 10.0.50.254

VLAN 60 (SIEM): 10.0.60.1/24 → VIP: 10.0.60.254

VLAN 70 (STORAGE): 10.0.70.1/24 → VIP: 10.0.70.254

VLAN 80 (PEERING): 10.0.80.1/24 → VIP: 10.0.80.254

```

## \*\*VLAN SVI IPs (Aggregation)\*\*

```

VLAN 10: 10.0.10.2/24 (Secondary .2)

VLAN 20: 10.0.20.2/24 (Secondary .2, with HSRP 10.0.20.253)

VLAN 30: 10.0.30.2/24

VLAN 40: 10.0.40.2/24

VLAN 50: 10.0.50.2/24

VLAN 60: 10.0.60.2/24

VLAN 70: 10.0.70.2/24

VLAN 80: 10.0.80.2/24

```

## \*\*Firewall IPs\*\*

```

FW1 Outside: 203.0.113.2/30

FW1 DMZ: 10.0.80.1/24

FW2 DMZ: 10.0.80.2/24

FW2 Inside: 10.0.50.254/24

\\\

Server IPs

\\\

Load-Balancer: 10.0.20.100/24 (Gateway: 10.0.20.254)

DB-SERVER: 10.0.30.100/24 (Gateway: 10.0.30.254)

SOC: 10.0.50.50/24 (Gateway: 10.0.50.254)

SIEM: 10.0.60.50/24 (Gateway: 10.0.60.254)

IDS: 10.0.50.51/24 (Gateway: 10.0.50.254)

IPS: 10.0.50.52/24 (Gateway: 10.0.50.254)

\\\

KEY CONFIGURATION CONSTANTS

OSPF Settings

\\\

Process ID: 1

Area: 0 (Backbone)

MD5 Key: BANKOSPF2026

\\\

BGP Settings

\\\

Bank AS: 65001

ISP-R1 AS: 100

ISP-R2 AS: 200

\\\

HSRP Settings

\\\

CORE-SW1 Priority: 110 (ACTIVE)

CORE-SW2 Priority: 100 (STANDBY)

AGG-SW1 Priority: 110 (ACTIVE on VLAN 20)

AGG-SW2 Priority: 100 (STANDBY on VLAN 20)

Preempt: Enabled on all

\\\

NAT Settings

\\\

Static NAT: 10.0.20.100 ↔ 203.0.113.150

PAT Network: 10.0.0.0/16 → WAN IP (overload)

^^
